

LECTURE 3

HTML and HTML5 Fundamentals

The raw ingredients of every website: structure, text, links, images, tables, and lists.

CSC336 Web Technologies

In This Lecture

- 1 Understand the role of markup languages, and HTML's document structure
- 2 Elements, tags, attributes, nesting rules, void elements, and validation
- 3 Headings, text formatting, and why the old `<font>` tag is obsolete
- 4 `<pre>/<code>` and other technical text tags, plus HTML entities
- 5 Hyperlinks (internal vs. external), images (including image links), audio, video
- 6 Tables, lists (including nested lists), and what HTML5 added

What Is a Markup Language?

- 1 HyperText — text that contains links to other text or pages
- 2 Markup — the text is annotated with tags describing its structure
- 3 Language — a shared grammar every browser agrees to follow
- 4 HTML is NOT a programming language — no variables, loops, or conditions; it only describes structure and content

STRUCTURE

HTML Document Structure

<!DOCTYPE html>, <html lang="en">, <head> (meta, title), and <body> — every valid page starts this way.

Hello, Web!

This is my first HTML page.

Void (Self-Closing) Elements

Element	What It Does
 	A single line break within text
<hr>	A thematic divider line between sections
	Embeds an image
<input>	A form control
<meta> / <link>	Document metadata / an external resource, used inside <head>

Void elements have no content and therefore no closing tag — <div></div> is NOT void, it just happens to be empty.

Void Elements, Rendered

A line break, a horizontal rule, a broken-image icon showing its alt text, and an empty input.

Text before a line break.

Text after the break.

A photo of a cat

Block-Level vs. Inline Elements

	Block-Level	Inline
Starts a new line?	Yes	No — flows with content
Full available width?	Yes, by default	No — only as wide as content
Examples	h1-h6, p, div, ul, table	a, strong, em, span, img

Block vs. Inline, Rendered

The heading and both paragraphs each claim a full line; strong and a sit inside the last paragraph's line.

Page Title

This is a paragraph.

This is another paragraph, on its own line below.

This word is **important**, and this is a [link](#) in the middle of a sentence.

Global Attributes

Attribute	Purpose
id	A unique identifier for one specific element
class	Category name(s) shared by many elements, for CSS/JS
title	Extra info shown as a tooltip on hover
style	Inline CSS applied to just this one element
lang	The (human) language of this element's content
data-*	A custom attribute for attaching your own data

Boolean attributes (required, checked, disabled) need no ="value" — present means true, absent means false.

Attributes, Rendered

A link, an abbr with a title tooltip, a style-colored paragraph, and three boolean-attribute form controls.

[Visit COMSATS](#)

Founded in 1998, W3C sets web standards.

Urgent



Subscribe

Headings and Paragraphs

<h1> through <h6>, most to least important — only one <h1> per page.

Chapter Title

Section Heading

Sub-section Heading

Text Formatting Elements

strong, em, b, i, mark, small, sub/sup, br, hr — each with a distinct default appearance.

Warning: this is *really* important.

Bold text and *Italic text*.

highlighted text, and terms apply.

H₂O and x²

Line one

Line two

A DEPRECATED TAG

The Tag

```
<font color="red" size="5" face="Arial">  
  This text is red, size 5, Arial font.  
</font>
```

Before CSS existed, was the only way to style text directly in HTML.

A DEPRECATED TAG

, Rendered

This text is red, size 5, Arial font.

Why We Don't Use

- Mixes structure with appearance — HTML should describe what something is, not how it looks
- Doesn't scale — changing 50 pages means editing 50 pages, not one CSS rule
- No responsive or interactive ability — can't change on hover, on a smaller screen, or in dark mode
- Fails validation — flagged as an error by the W3C Markup Validator

<pre>, <code>, and Related Tags

Tag	Meaning
<pre>	Preformatted text — preserves every space and line break, monospace
<code>	An inline snippet of computer code, monospace
<kbd>	Keyboard input — a key the user should press
<samp>	Sample output from a program or command
<var>	A variable name or placeholder value
<blockquote>	A longer, block-level quotation

HTML Entities

Entity	Renders As	Why You Need It
&lt;	<	A literal < would start a tag
&gt;	>	A literal > would end a tag
&amp;	&	A literal & would start another entity
&nbsp;	(a space)	A non-breaking space — never wraps a line
&copy;	©	Not on most keyboards
&mdash;	—	An em dash, longer than a hyphen

Internal and External Links

EXTERNAL (ABSOLUTE URL)

- The complete address, including https:// and the domain
- Use for a page on a DIFFERENT website
- `target="_blank" + rel="noopener noreferrer"` opens safely in a new tab

INTERNAL (RELATIVE URL)

- A path relative to the current page — no domain name
- Use for pages within YOUR OWN site
- Survives a domain move, since it never contained the domain

Links, Rendered

External links first (Wikipedia, COMSATS in a new tab), then internal ones (About, Contact, Home).

[HTML on Wikipedia](#)

[Visit COMSATS \(new tab\)](#)

[About Us](#)

[Contact \(in a subfolder\)](#)

[Back to home \(one folder up\)](#)

The `<img>` Element

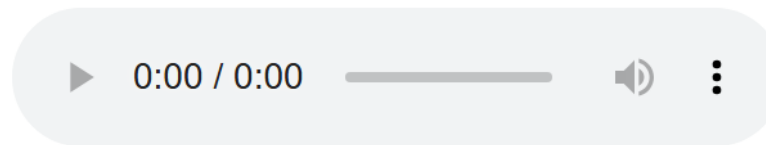
src, alt, width, height — and wrapping an `<img>` in `<a>` turns it into a clickable image link.

A dark blue rectangular box with the word "COMSATS" in white, bold, uppercase letters centered inside.

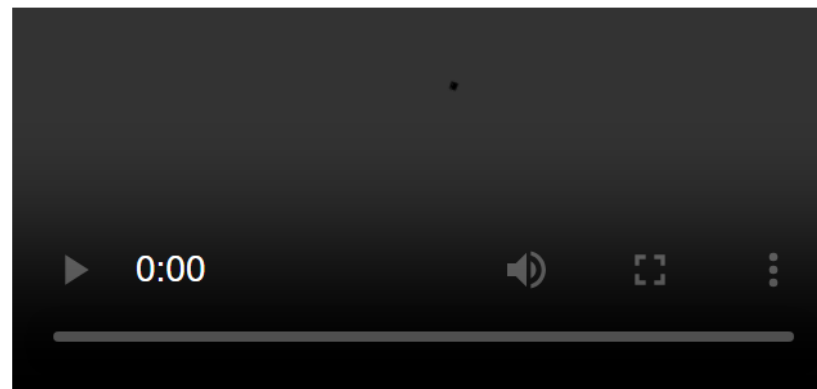
Audio and Video

<audio controls> and <video controls> with a <source> child — native players, no plugins needed.

Audio player:



Video player:



Tables

`<table>`, `<thead>/<tbody>`, `<tr>`, `<th>`, `<td>` — for genuinely tabular data only, never page layout.

Name	Course	Grade
Ali	CSC336	A
Sara	CSC336	A+

Nested Lists

```
<ul>
  <li>Front End
    <ul>
      <li>HTML</li>
      <li>CSS</li>
    </ul>
  </li>
  <li>Back End</li>
</ul>
```

A nested `<ul>/<ol>` goes inside the `<li>` of the item it belongs under. Browsers change bullet style (disc, circle, square) at each depth.

All Three List Types, Rendered

Unordered (bulleted), ordered (numbered), and description (term/definition pairs).

- HTML
- CSS
- JavaScript

1. Write the HTML
2. Add CSS styling
3. Add JavaScript behavior

HTML

The markup language used to structure web pages.

CSS

The language used to style web pages.

What HTML5 Added

- Semantic layout: header, nav, main, section, article, aside, footer
- Media: audio, video, source, track — no plugins required
- Graphics: canvas and native svg support
- New attributes: placeholder, required, pattern, data-*, contenteditable, draggable

Progressive Enhancement

TIP

Not every browser supports every HTML5 feature. Check caniuse.com before relying on a newer one, and design your page so it still works reasonably well in older browsers while enhancing the experience for those that support more.

KEY TAKEAWAYS

Lecture 3 in Six Points

- 1 HTML describes structure and content only — void elements (`br`, `img`) have no closing tag; nesting must close in reverse order.
- 2 Block-level elements stack and fill width; inline elements flow within text.
- 3 The `<font>` tag is obsolete — use CSS for all styling instead.
- 4 `<pre>/<code>` handle technical text; entities (`<`, ` `) represent characters HTML can't take literally.
- 5 External links use absolute URLs; internal links use relative ones and survive a domain move.
- 6 HTML has three list types — unordered, ordered, description — and any of them can be nested.